

- Description

This module manages a smart gas leak detection system, ensuring that:

1. **Gas concentration levels are continuously monitored to detect leaks in a timely manner.**
2. **If the gas concentration exceeds a predefined safety threshold, the gas supply is automatically shut off.**
3. **If the gas concentration returns to a safe level, the system allows manual reactivation of the gas supply.**
4. **A log of all gas level readings, emergency shutdowns, and system reactivations is maintained for safety compliance.**

This system helps prevent gas-related hazards, enhance household and industrial safety, and ensure rapid response to leaks.

- Specification

module SmartGasLeakDetection

var currentGasLevel, maxSafeLevel: real

var gasSupplyActive: boolean

process updateGasLevel(changeAmount: real)

functional scenario form:

currentGasLevel $\leq$ maxSafeLevel && Gas concentration updated successfully.

||

currentGasLevel > maxSafeLevel && gasSupplyActive = false, Gas leak detected, gas supply shut off.

end process

process resetGasSupply()

functional scenario form:

(currentGasLevel $\leq$ maxSafeLevel) && gasSupplyActive = true, Gas supply reactivated.

OR

(currentGasLevel > maxSafeLevel) && gasSupplyActive = false, Gas supply remains shut off.

end process

end module

- Java Code

```
package code;
class SmartGasSystem {
    private double maxSafeLevel;
    private double currentGasLevel;
    private boolean gasSupplyActive;
    public SmartGasSystem(double maxSafeLevel, double initialGasLevel)
{
}
```

```

        this.maxSafeLevel = maxSafeLevel;
        this.currentGasLevel = initialGasLevel;
        this.gasSupplyActive = true;
    }
    public void updateGasLevel(double changeAmount) {
        currentGasLevel += changeAmount;
        manageGasSupply();
        System.out.printf("Current Gas Concentration: %.2f%%\n", currentGasLevel);
        if (!gasSupplyActive) {
            System.out.println("WARNING: Gas Leak Detected! Gas Supply Shut Off.");
        }
    }
    private void manageGasSupply() {
        if (currentGasLevel > maxSafeLevel) {
            gasSupplyActive = false;
        }
    }
    public void resetGasSupply() {
        if (currentGasLevel <= maxSafeLevel) {
            gasSupplyActive = true;
            System.out.println("Gas Supply Reactivated.");
        } else {
            System.out.println("Cannot Reactivate Gas Supply. Gas Concentration Still Unsafe.");
        }
    }
}

public class c62_SmartGasLeakSystem {
    public static void main(String[] args) {
        SmartGasSystem gasSystem = new SmartGasSystem(5.0, 2.0);

        // testcase-VT:
        gasSystem.updateGasLevel(2.5);
        gasSystem.updateGasLevel(1.0);
        gasSystem.resetGasSupply();
        gasSystem.updateGasLevel(-2.0);
        gasSystem.resetGasSupply();
        // testcase-FT:
        gasSystem.updateGasLevel(1.7976931344486818E308);
        gasSystem.updateGasLevel(-4.9E-324);
        gasSystem.updateGasLevel(0.0);
        gasSystem.updateGasLevel(-1.7976931348623157E308);
    }
}

```

```
//      gasSystem.updateGasLevel(9.094212329303478E307);

//      // testcase-Z3:
//      gasSystem.updateGasLevel(2.5);
//      gasSystem.updateGasLevel(1.0);
//      gasSystem.resetGasSupply();
//      gasSystem.updateGasLevel(-2.0);
//      gasSystem.resetGasSupply();
//      // testcase-UVT:
//      gasSystem.updateGasLevel(2.5);
//      gasSystem.updateGasLevel(1.0);
//      gasSystem.resetGasSupply();
//      gasSystem.updateGasLevel(-2.0);
//      gasSystem.resetGasSupply();
//  }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c62_SmartGasLeakSystem.java	100.0 %	6	0	6
> c62_SmartGasLeakSystem		0	0	0
> SmartGasSystem	100.0 %	6	0	6

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,642	20,728
▼ src	0.4 %	86	20,642	20,728
▼ code	0.4 %	86	20,642	20,728
▼ c62_SmartGasLeakSystem.java	96.6 %	86	3	89
> c62_SmartGasLeakSystem	87.0 %	20	3	23
> SmartGasSystem	100.0 %	66	0	66

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	839	842
▼ src	0.4 %	3	839	842
▼ code	0.4 %	3	839	842
▼ c62_SmartGasLeakSystem.java	50.0 %	3	3	6
> c62_SmartGasLeakSystem		0	0	0
> SmartGasSystem	50.0 %	3	3	6

5. Fuzz testing instruction coverage: 78%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	71	20,659	20,730
▼ src	0.3 %	71	20,659	20,730
▼ code	0.3 %	71	20,659	20,730
▼ c62_SmartGasLeakSystem.java	78.0 %	71	20	91
> c62_SmartGasLeakSystem	88.0 %	22	3	25
> SmartGasSystem	74.2 %	49	17	66

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c62_SmartGasLeakSystem.java	100.0 %	6	0	6
> c62_SmartGasLeakSystem		0	0	0
> SmartGasSystem	100.0 %	6	0	6

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,642	20,728
▼ src	0.4 %	86	20,642	20,728
▼ code	0.4 %	86	20,642	20,728
▼ c62_SmartGasLeakSystem.java	96.6 %	86	3	89
> c62_SmartGasLeakSystem	87.0 %	20	3	23
> SmartGasSystem	100.0 %	66	0	66

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c62_SmartGasLeakSystem.java	100.0 %	6	0	6
> c62_SmartGasLeakSystem		0	0	0
> SmartGasSystem	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,642	20,728
▼ src	0.4 %	86	20,642	20,728
▼ code	0.4 %	86	20,642	20,728
▼ c62_SmartGasLeakSystem.java	96.6 %	86	3	89
> c62_SmartGasLeakSystem	87.0 %	20	3	23
> SmartGasSystem	100.0 %	66	0	66